5 Bash Scripts Smart DBAs Use to Win Back Their Time

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Here are five scripts that quietly turn grind into glide:

Tablespace-Growth Watchdog: Spots space creep early so production never gasps for air.

Archive Log Purger: Keeps log volumes lean, sparing you late-night disk alarms.

Backup-Integrity Verifier: Validates backups in the background so restores never surprise you.

Index-Fragmentation Reporter: Flags performance-killing bloat before users even notice.

Password-Expiry Notifier: Gives teams fair warning and keeps audits calm.
For each script you get:

1. **What it does and why it matters**
2. The **full shell script** (copy-ready)

1. Tablespace-Growth Watchdog

What it does

Checks every Oracle tablespace once per hour. If any free space drops below 10 %, it sends an alert and writes a log.

Script: tblspace_watchdog.sh

```
#!/usr/bin/env bash
# Tablespace-Growth Watchdog

ORACLE_SID=ORCL
export ORACLE_SID
export ORAENV_ASK=NO
. oraenv >/dev/null

THRESHOLD=10   # percent free
LOG_DIR=/var/log/tblspace_watchdog
mkdir -p "$LOG_DIR"

sqlplus -s / as sysdba <<'SQL' > "$LOG_DIR/space_$(date +%F_%H%M).log"
```

```
set pages 0 feedback off heading off
SELECT tablespace_name || '|' ||
      ROUND((free_space/total_space)*100) pct_free
FROM   (SELECT a.tablespace_name,
               b.total_space,
               a.free_space
        FROM   (SELECT tablespace_name,
                       SUM(bytes)/1024/1024 free_space
                FROM   dba_free_space
                GROUP  BY tablespace_name) a,
               (SELECT tablespace_name,
                       SUM(bytes)/1024/1024 total_space
                FROM   dba_data_files
                GROUP  BY tablespace_name) b
        WHERE  a.tablespace_name = b.tablespace_name);
SQL

while IFS='|' read -r tbs pct; do
  if [[ "$pct" -lt "$THRESHOLD" ]]; then
    echo "Tablespace $tbs only ${pct}% free" \
    | mailx -s "[TBS-Alert] $tbs low space" dba_team@example.com
  fi
done < <(tail -n +1 "$LOG_DIR"/space_*.log | grep '|')
```

2. Archive Log Purger

What it does

Finds archive logs older than 7 days on each Oracle host and deletes them safely, reclaiming disk space.

Script: archive_purger.sh

```
#!/usr/bin/env bash
HOSTS=("db1" "db2")
AGE_DAYS=7
for H in "${HOSTS[@]}"; do
  ssh "$H" "
    find /u02/archivelog -type f -mtime +$AGE_DAYS -name '*.arc' -print -delete \
    > /u02/archivelog/purge_\$(date +%F).log 2>&1"
done
```

3. Backup-Integrity Verifier

What it does

Runs RMAN VALIDATE on the most recent full backup, captures output, and alerts if corruption appears.

Script: backup_verify.sh

```
#!/usr/bin/env bash
ORACLE_SID=ORCL
export ORACLE_SID
export ORAENV_ASK=NO
. oraenv >/dev/null

LOG=/var/log/backup_verify_$(date +%F).log
rman target / <<RMAN > "$LOG"
crosscheck backup;
validate backupset tag 'FULL_BACKUP';
exit
RMAN
```

```
if grep -qi "Corrupt" "$LOG"; then
  mailx -s "[Backup-Alert] Corruption detected" dba_team@example.com < "$LOG"
fi
```

4. Index-Fragmentation Reporter

What it does

Lists Oracle indexes whose **leaf-block waste exceeds 20 percent**, writes the result to a CSV file, and emails the team so they can schedule a rebuild or coalesce.

Script: ora_index_bloat_report.sh

```
#!/usr/bin/env bash
# Oracle Index-Fragmentation Reporter

ORACLE_SID=ORCL
export ORACLE_SID
export ORAENV_ASK=NO
. oraenv >/dev/null          # Sets Oracle env on most Unix hosts

THRESHOLD=20                 # percent leaf-block bloat
CSV=/tmp/idx_bloat_$(date +%F).csv

sqlplus -s / as sysdba <<SQL > "$CSV"
SET pages 0 feedback off heading off
-- Calculate leaf-block waste: (leaf_blocks - DISTINCT_KEYS) / leaf_blocks
SELECT owner || '.' || index_name || ',' ||
       leaf_blocks || ',' ||
       DISTINCT_KEYS || ',' ||
       ROUND((leaf_blocks - DISTINCT_KEYS)*100/leaf_blocks,2) AS pct_bloat
```

```
FROM   dba_indexes
WHERE  status = 'VALID'
  AND  leaf_blocks > 0
  AND  (leaf_blocks - DISTINCT_KEYS)*100/leaf_blocks > $THRESHOLD
ORDER  BY pct_bloat DESC;
SQL

if [[ -s "$CSV" ]]; then
  mailx -s "[IDX-Bloat] Indexes over ${THRESHOLD}% leaf waste" \
        -a "$CSV" dba_team@example.com < /dev/null
fi
```

5. Password-Expiry Notifier

What it does

Checks Oracle accounts for passwords that expire in the next 7 days and emails a list so users can reset early.

Script: pwd_expiry_notifier.sh

```
#!/usr/bin/env bash
ORACLE_SID=ORCL
export ORACLE_SID
export ORAENV_ASK=NO
. oraenv >/dev/null

THRESHOLD=7
OUT=/tmp/pwd_expiry_$(date +%F).txt

sqlplus -s / as sysdba <<SQL > "$OUT"
set lines 120
```

```
col username format a20
SELECT username, expiry_date
FROM dba_users
WHERE account_status='OPEN'
  AND expiry_date BETWEEN sysdate AND sysdate + $THRESHOLD
ORDER BY expiry_date;
SQL

if [[ -s "$OUT" ]]; then
  mailx -s "[Password-Alert] Accounts expiring soon" dba_team@example.com < "$OUT"
fi
```

Final checklist

1. Confirm each script's paths, hostnames, email addresses, and Oracle environment variables fit your shop.
2. Verify password-less SSH for scripts that reach other hosts.
3. Keep logs rotating with logrotate or by adding find … -delete lines.

Implementing these five automations can reclaim several hours every week by replacing manual checks with silent, condition-based alerts.